COS40003: Concurrent Programming

Report 1: Learning Summary (Weeks 1–3)

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Week 1 - Introduction

The summary below for Week 1 covers **Part I** of **Lab 1**.

1.1 Concurrent Computing

Concurrent computing is a fundamental computing paradigm that means multiple computational tasks are being executed on a same time period, but not necessarily simultaneously. Instead of processing tasks sequentially, the OS (operating system) will manages several execution flows at once to maximize resource utilization.

Concurrent computing relies heavily on **time-sharing**. The OS divides the CPU into small execution flows and rapidly switches between active processes. This approach fundamentally maximizes system effciency by allowing processes that are ready to be executed to be processed continously.

Coordinating concurrent tasks requires addresing two fundamental challanges:

1. **Scheduling:** Determining which process executes next to ensure fairness and response time.
2. **Concurrency Control:** Managing shared resources to prevent **race conditions** (inconsistent data or state caused by timing issues) and **deadlocks** (processes that wait indefinitely where two or more threads or processes wait indefinitely for each other, which prevents any of them from making progress).

A common example of concurrent computing is a web browser (e.g., Chrome, Safari) is opening multiple tabs at one, where each tab are doing seperate tasks such as playing a video, downloading a file, or fetching live chat messages.

According to Rob Pike, co-creator of the Go programming language: **"concurrency is about dealing with lots of things at once, while parallelism is about doing lots of things at once"** [1].

1.2 Parallel Computing

Parallel computing is a computing architecture in which multiple calculations or execution flows are carried out at the exact same time. Concurrent computing often creates an illusion of simultaneous progress on a single CPU from the user's perspective. Parallelism, on the other hand, represents multiple executions at the exact same time, as illustrated in Figure 1.

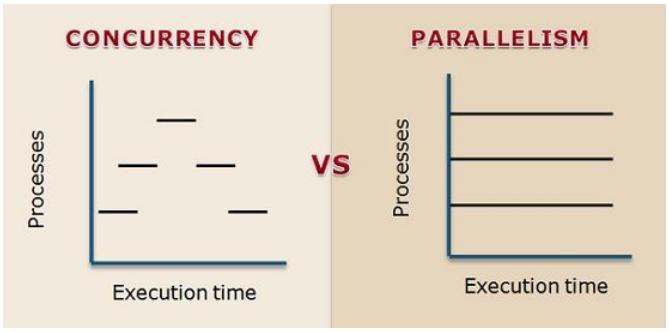


Figure 1: Concurrent execution vs. Parallel execution [2].

In parallel computing, a large computational workload is broken down into smaller, independent subtasks that can be executed concurrently. Each subtasks are executed simultaneously on seperate processors, and once they are finished, the individual sub-results are merged to produce the final output. In a typical parallel computing architectures, processors communicate directly through **shared memory** to ensure low-latency data exchange across processes.

A real-world example of parallel computing is a **Graphics Processing Unit (GPU)**, which comprises thousands of smaller arithmetic cores for executing identical operations such as rendering graphics or computing matrix multiplications using the **Single Instruction, Multiple Data (SIMD) architecture** (shown in Figure 2).

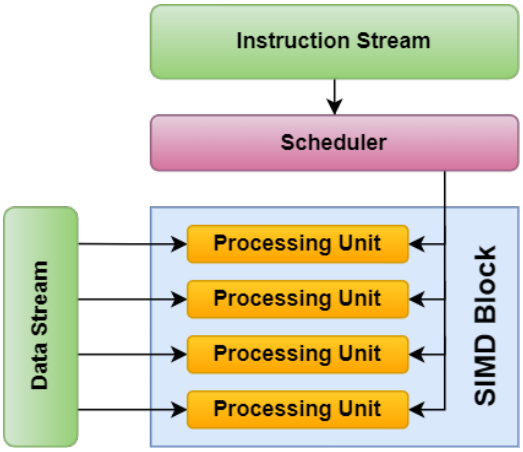


Figure 2: SIMD Architecture [3].

1.3 Distributed Computing

Distributed computing consists of multiple autonomous computing entities that are connected through a shared network to solve complex computational problems. A fundamental architectural distinction between parallel computing and distributed computing lies in memory organization. In parallel computing, processors shares a common memory space to exchange information directly. But for distributed computing, each individual computer cannot access each other's memory directly and must exchange information through **message passing** (communication protocols or middleware) [4] (shown in Figure 3).

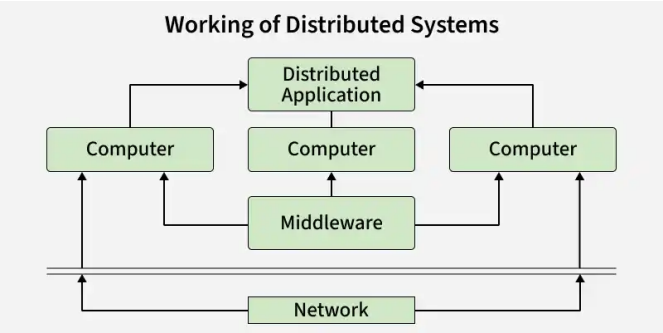


Figure 3: Architecture of a Distributed System [4].

Distributed computing architectures are commonly classified into two primary paradigms:

1. **Cluster computing:** Composed of **homogeneous** machines, which feature similar hardware configurations and identical OSs, that are **co-located** within a single area. This architecture is designed for high availability and high-performance computing.
2. **Grid computing:** Comprised of **heterogeneous** computing entities that are **geographically dispersed** across different physical locations. Grid computing represents the conceptual predecessor to modern cloud infrastructure.

A real-world example that implements both cluster and grid computing is the **Worldwide LHC Computing Grid (WLCG)** [5]. Local server clusters are deployed for high-performance batch processing, and these clusters are linked globally into a centralized grid to analyze high volumes of data.

1.4 Cloud Computing

Cloud Computing is one of the most demanded computational infrastruture in modern computing, delivering services such as servers, storage, databases, and networking over the Internet.

The fundamental goal of cloud computing is abstraction, which hides the underlying physical hardware and maintenance complexity from users, allowing them to access on demand resources with seamless scalability and minimal configuration. Behind this abstraction, cloud computing relies extensively on distributed systems since it coordinates massive networks of server clusters to ensure high availability and data redundancy across multiple regions.

Cloud services are organized into three primary architectural models:

1. **Infrastructure as a Service (IaaS):** Provides virtualized computing infrastructure, including virtual machines, block storage, and virtual networks (e.g., AWS EC2, Microsoft Azure VMs)
2. **Platform as a Service (PaaS):** Provides a managed execution environment and development framework where developers can build and run applications without configuring the underlying OS or runtime (e.g., Vercel, Firebase)
3. **Software as a Service (SaaS):** Provides software applications that can be directly accessed via web browsers (e.g., Microsoft 365, Google Docs).

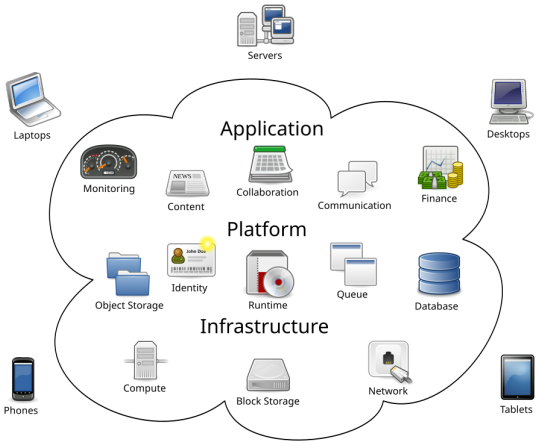


Figure 4: Cloud Computing Logical Architecture and Service Models [6].

1.5 Fog and Edge Computing

Fog and Edge computing designed to bring processing and data storage closer to data-generating sources, primarily Internet of Things (IoT) devices. These computing paradigms address the limitations of address critical limitations of cloud computing, such as transmission latency and bandwidth bottlenecks.

- **Edge Computing:** Performs computation directly on local hardware devices such as microcontrollers, sensors, and local gateways.
- **Fog Computing:** Extends the cloud computing paradigm toward the physical local environment by introducing an intermediate layer that coordinates compute resources between local devices (nodes) and the remote cloud.

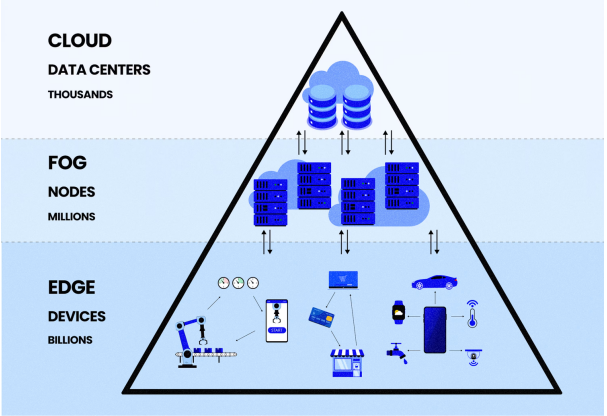


Figure 5: Edge Computing Architecture and Data Processing Flow [8].

A real-world implementation that utilizes both paradigms is smart traffic management in intelligent transportation systems [7]. Vehicle sensors process immediate collision data locally at the edge devices, while local base station units operate as an intermediate fog layer nodes to coordinate traffic flow and signal timings in real time.

Lab 1 - Part II

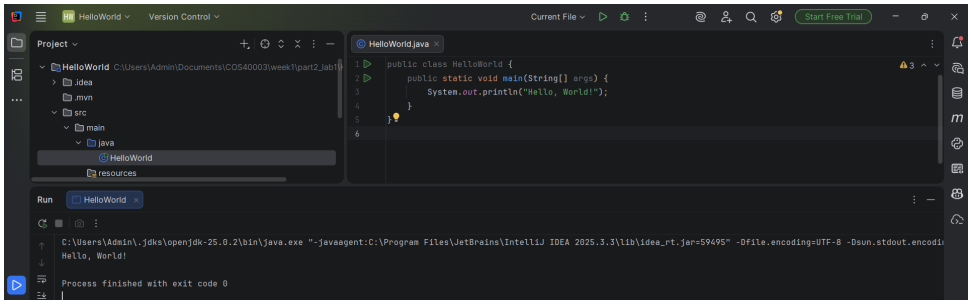


Figure 6: The "Hello World!" Java application compiled and ran successfully.

Week 2 - Process: Concept and API

The summary below for Week 2 covers **Part I** of **Lab 2**.

2.1 Process Definition

A process is essentially a program that being executed. When a process is running, the OS allocates an address space in RAM (containing code, data, heap and stack) and the CPU read the instructions and execute them on by one.

The differences between a process and a program lies in their execution state. A program is stored as an executable binary on the disk, wheras a process is an active instance executing dynamically in memory.

2.2 The Need for Processes

Early computing executed single programs one at a time, which creates severe bottlenecks whenever a task had to wait for slow I/O operations, which forces the processor to idle. The abstraction of process was invented to solve this inefficiency, allows the OS to safely execute multiple programs concurrently with isolated memory.

The famous computer scientist Edsger W. Dijkstra has demonstrated the necessity of process through showing that breaking complex computing workloads into separated sequential execution units is essential to keep the CPU continuously utilized while preventing tasks from overwriting shared data [10].

2.3 Process Operations

2.3.1 Process Components

A process comprises of four essential elements:

- **Image of a program:** The compiled code, environment, and dependencies.
- **Memory (Address Space):** An isolated virtual address space allocated to each running process (shown in Figure 7). The program instructions data and static data remain fixed throughout execution, while two dynamic memory segments grow towards each other:

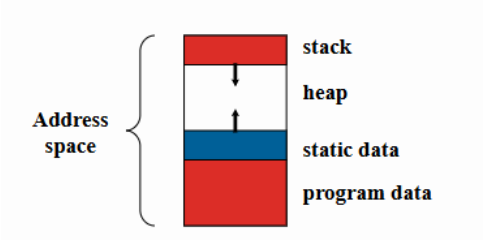


Figure 7: Process Address Space Structure [9].

  - **Stack:** Managed automatically by the CPU in Last-In, First-Out (LIFO) order to store variables, function parameters, and return addresses. It grows downward toward lower memory addresses.
  - **Heap:** Managed at runtime for for dynamic memory allocation. It grows upward toward higher memory addresses.

- **CPU state:** Registers, Program Counter (PC), and Stack Pointer (SP).
- **OS state:** Metadatas such as opened files, temporal statistics, etc.

2.3.2 Context Switch

Context switch is a mechanism that alternates the CPU between different processes to enable multitasking. Because a single CPU core can execute only one process at a time, the OS rapidly switches active execution contexts between processes (represented as programs `Prog1` and `Prog2` in Figure 8) across four phases:

1. **Phase 1 (`Prog1` Running):** `Prog1` executes instructions on the CPU, while `Prog2` remains suspended.
2. **Phase 2 (Save State):** The OS saves the CPU registers, PC, SP of `Prog1`, placing it into suspension.
3. **Phase 3 (Restore State):** The OS restores the previously saved registers, PC, and SP belonging to `Prog2`.
4. **Phase 4 (`Prog2` Running):** `Prog2` takes over the CPU and resumes execution, while `Prog1` waits in memory.

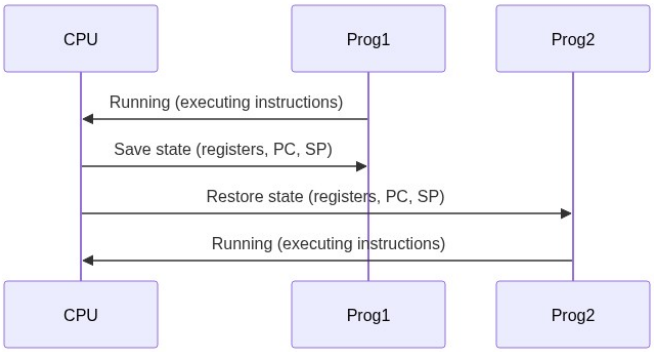


Figure 8: Process Context Switching Mechanism Between Two Processes.

2.3.3 The Process Abstraction

The **process abstraction** is a conceptual bridge between shared physical hardware and private program execution processes:

- **Physical View:** One physical CPU shared and context-switched among processes.
- **Logical View:** Each process operates as if it has its own dedicated CPU and memory.
- **Time-Sharing:** Processes rapidly take turns using the CPU, enabling multi-computational processes on a single core.

2.3.4 Process States

A process transitions through three primary execution states:

- **Ready:** The process is loaded in memory and prepared to run.
- **Running:** The process currently holds the CPU and is actively executing instructions.
- **Blocked:** The process has released control of the CPU to wait for an event (such as I/O completion) and cannot execute until that event happens.

Transitions between these states occur as follows:

- **Ready → Running:** The scheduler picks the process to run.
- **Running → Ready:** The scheduler suspends the process.
- **Running → Blocked:** The process waits for an event (e.g., I/O request).
- **Blocked → Ready:** The event occurs (e.g., I/O completes).

2.4 Process APIs

There are three fundamental process management APIs in Unix systems.

1. **`fork()` - Creating a Process by Duplication:**

   Creates an exact clone of the caller with its own memory space. It returns `0` to the child process and the child's Process ID (PID) to the parent, allowing both to branch into different execution paths.

2. **`wait()` - Synchronizing with Child Termination:**

   Pauses the calling parent until one of its child processes finishes execution. It returns the terminated child's PID and collects its exit status, preventing zombie processes (dead processes waiting for their parents).

3. **`exec()` - Replacing Current Program Code:**

   Replaces the process's current memory contents (code, data, heap, stack) with a new executable program from disk. The process retains its original PID but executes the new code.

Lab 2 - Part II

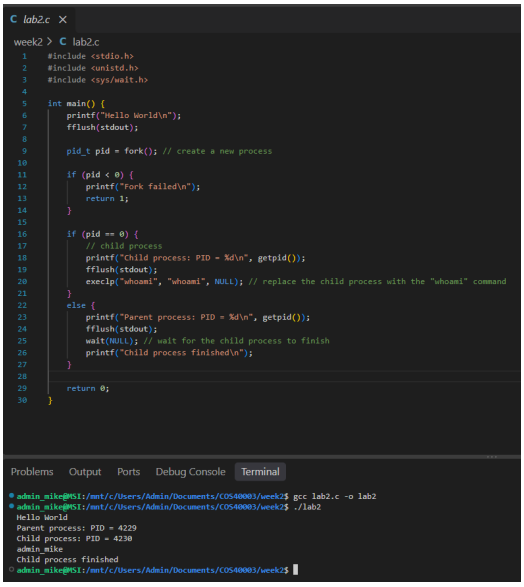


Figure 9: Example Hello World C Program using `fork()`, `wait()`, and `exec()` APIs.

Week 3 - Scheduling

The summary below for Week 3 covers **Part I** of **Lab 3**.

3.1 CPU Scheduling

The OS must decide which process runs next on the CPU. This decision directly impacts system performance and the user experience.

There are several **CPU Scheduling Criterias** that the OS can use:

- **CPU Utilization:** The total time of the CPU being busy executing processes.
- **Turnaround Time:** The total time of a job submission to its termination.
- **Response Time:** The amount of time it takes to get the first response.
- **Missed Deadlines:** The number of jobs that fail to complete before their deadlines.

Beyond these performance criteria, there are several **CPU Scheduler Options** that define how scheduling policies are governed:

- **Priorities:** Determines which process will be executed based on its priority level.
- **Preemptive vs. Non-preemptive:**
  - **Preemptive:** The OS can interrupt a running task at any time to take over.
  - **Non-preemptive:** The OS will not interrupt a running task until it finishes its execution.

3.2 First-Come, First-Served (FIFO)

The **First Come, First-Served algorithm** takes the first process that arrives and executes it until completion. Processes that comes next will wait in a queue until the CPU is avaliable (FIFO queue).

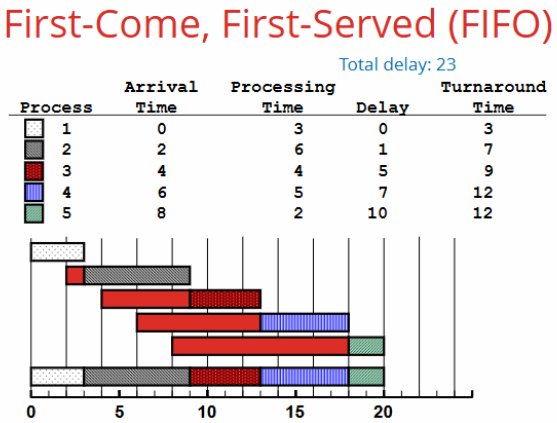


Figure 10: FCFS Scheduling Example [11].

As shown in Figure 10, five processes are scheduled non-preemptively based on their arrival times. Process 1 arrives at $t = 0$ and runs immediately, while subsequent processes must wait in the ready queue until the running task completes. Shorter processes that arrive later suffer significant waiting time such as Process 5, which requires only 2 units of processing time but experiences a delay of 10 units, which results in a high cumulative delay of 23 units.

3.3 Shortest Job First (non-preemptive)

The **Shortest Job First (SJF) algorithm** selects the process with the shortest estimated execution time from the ready queue. This approach is non-preemptive, so that once a process starts executing, it will run to completion before the next process is selected.

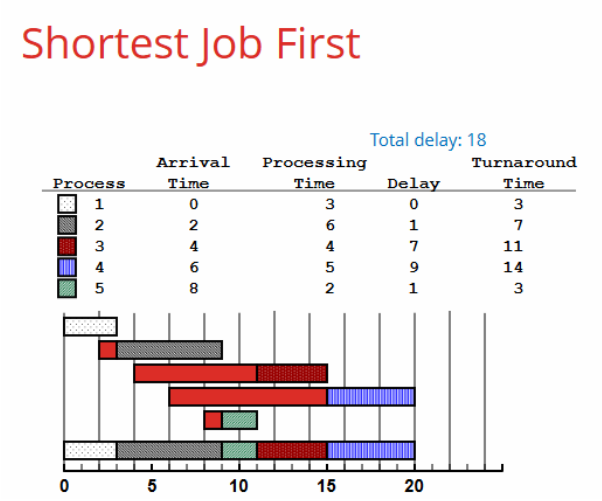


Figure 11: SJF Scheduling Example [11].

As shown in Figure 11, after Process 1 and Process 2 complete at time unit 9, Process 5 is scheduled next because its required processing time is only 2 units, compared to 4 units for Process 3 and 5 units for Process 4. Prioritizing shorter jobs significantly decreases waiting time, bringing the total delay down to only 18 units.

3.4 Shortest Job First (with preemption)

The **Shortest Job First with preemption** also known as **Shortest Remaining Time (SRT)**. Whenever a new process arrives in the ready queue, the scheduler compares its total processing time against the remaining time of the currently running task. If the new process has a shorter remaining burst time, the OS will preempt the running process by moving it back to the ready queue, then immediately allocates the CPU to the new process.

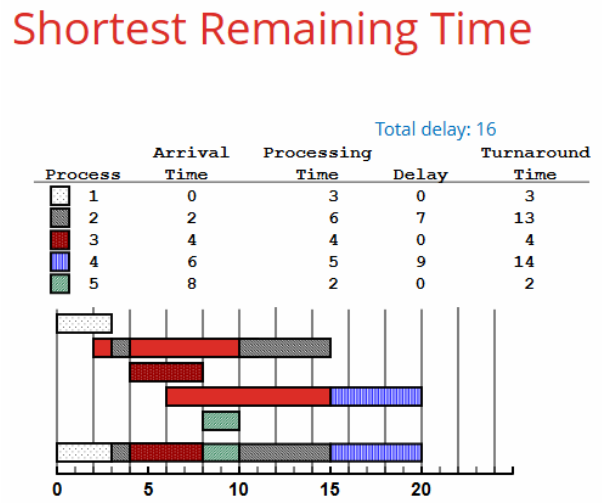


Figure 12: SRT Scheduling Example [11].

As shown in Figure 12, at time unit 4, Process 3 preempts Process 2 because its required 4 units of processing time is shorter than the 5 units remaining for Process 2. Prioritizing shorter jobs immediately upon arrival allows Process 3 and Process 5 to finish with zero delay, reducing the total delay to 16 units.

3.5 Round-Robin Scheduling

The **Round-Robin (RR) Scheduling algorithm** limits the amount of time a process can run on the CPU to a fixed time slice (quantum). If the process does not finish within its allocated time, it is preempted for the next process in the queue. This approach ensures that all processes receive fair amount of CPU time.

The effectiveness of Round-Robin Scheduling depends on the the number of jobs and the size of the time quantum. Large number of jobs means that the time between scheduling of a single job increases, which inherently increases the waiting time. While larger time quantum means that the time between the scheduling of a single job also increases, which also increases the waiting time. However, smaller time quantum would leads to better response time but also increases overhead due to more frequent context switches.

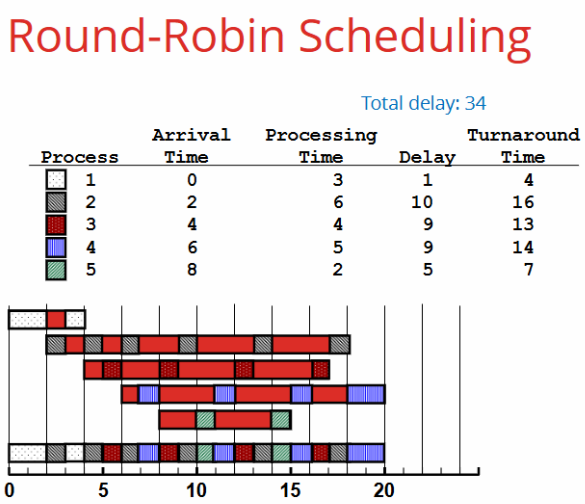


Figure 13: Round-Robin Scheduling Example [11].

As shown in Figure 13, with a time quantum of 1 unit, the CPU continuously cycles through active jobs in the queue to grant each process immediate execution. Because execution is fragmented across different time slices, longer processes take much longer to reach final completion, which results in a high cumulative delay of 34 units.

3.6 Lottery Scheduling

The **Lottery Scheduling algorithm** is a type of proportional share scheduling in which the scheduler assigns a certain number of lottery tickets to each process. To select the next process to run, the scheduler randomly generates a winning ticket number, and the process holding the matching ticket receives CPU execution time.

For example, if there are 100 total tickets and Process A holds 50 tickets, Process B holds 15 tickets, and Process C holds 35 tickets, then Process A has a 50% chance of being selected, Process B has a 15% chance, and Process C has a 35% chance.

3.7 Multi-Level Feedback Queue (MLFQ)

The **Multi-Level Feedback Queue (MLFQ) algorithm** is designed to achieve two goals at once:

1. **Optimize Turnaround Time:** Algorithms like SJF or SRT are great for this, but we don't know the job lengths.
2. **Minimize Response Time:** Algorithms like RR are great for this, but produce terrible turnaround time.

MLFQ uses multiple queues with different priority levels. There are several rules that govern how processes are scheduled and moved between queues:

Rule 1: If Priority(A) > Priority(B), A runs and B doesn't.

Rule 2: If Priority(A) = Priority(B), A and B run in RR.

Rule 3: When a job enters the system, it is placed at the highest-priority queue.

Rule 4a: If a job uses up an entire time slice while running, its priority is reduced.

Rule 4b: If a job gives up the CPU before the time slice is up, it stays at the same priority level.

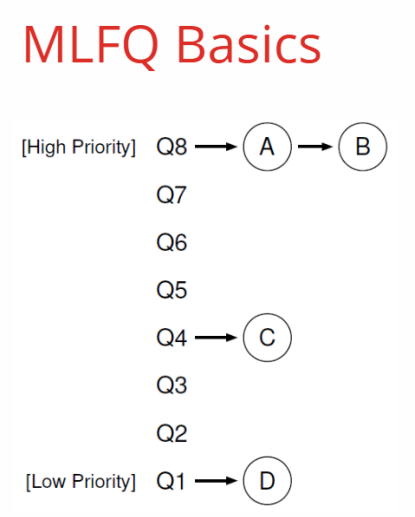


Figure 14: Multi-Level Feedback Queue Example [11].

As shown in Figure 14, processes are distributed across priority tiers from Q8 to Q1. The scheduler selects jobs from the highest queue, executing Process A and Process B in RR at Q8 while Process C (at Q4) and Process D (at Q1) wait.

Lab 3 - Part II

What are the main tasks of the scheduler? Allocating CPU time to jobs, measuring time slices, assigning job priorities, and preempting running tasks when necessary.

What is the difference between preemptive and non-preemptive scheduling? Preemptive scheduling allows the operating system to actively interrupt and suspend running processes, while non-preemptive scheduling allows processes to run until they complete.

What is the advantage of a shorter scheduling quantum? It reduces response time because there is a shorter delay time between execution turns.

What is the advantage of a longer scheduling quantum? It results in fewer context switches, which reduces system overhead.

Why is feedback scheduling useful for interactive jobs? Interactive tasks are usually much shorter and frequently start and stop. Because feedback scheduling assigns every new task the highest priority, interactive jobs are run at the highest priority.

Are these scheduling policies subject to starvation?

- **Shortest Job First scheduling?** Yes, longer jobs can starve if shorter jobs continuously arrive.
- **Round Robin scheduling?** No, every job is guaranteed CPU time within a bounded time quantum cycle.
- **First Come First Served scheduling?** No, jobs are processed in strict arrival order, so every job will eventually execute.
- **Fixed Priority scheduling?** Yes, lower-priority jobs will starve if higher-priority jobs keep occupying the CPU.
- **Multi-level feedback scheduling?** No, because priority boost moves all jobs back to the highest-priority queue periodically.

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